

Halogenated Metal–Organic Framework Glasses and Liquids

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Cite This: *J. Am. Chem. Soc.* 2020, 142, 3880–3890

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ABSTRACT: The synthesis of four novel crystalline zeolitic imidazolate framework (ZIF) structures using a mixed-ligand approach is reported. The inclusion of both imidazolate and halogenated benzimidazolate-derived linkers leads to glass-forming behavior by all four structures. Melting temperatures are observed to depend on both electronic and steric effects. Solid-state NMR and terahertz (THz)/far-IR demonstrate the presence of a Zn–F bond for fluorinated ZIF glasses. In situ THz/far-IR spectroscopic techniques reveal the dynamic structural properties of crystal, glass, and liquid phases of the halogenated ZIFs, linking the melting behavior of ZIFs to the propensity of the ZnN_4 tetrahedra to undergo thermally induced deformation. The inclusion of halogenated ligands within metal–organic framework (MOF) glasses improves their gas-uptake properties.



■ INTRODUCTION

Current research on metal–organic frameworks (MOFs) has concentrated predominantly on the properties of ordered crystalline phases.^{1,2} However, there is growing recognition of the importance of the physical properties of MOFs. In particular, the role of disorder, defects, and structural flexibility in installing beneficial physical behavior is now widely studied.^{3–8} In fact, it has recently been demonstrated that crystallinity itself is not a prerequisite for many of the properties for which MOFs are well-known, such as porosity, catalytic activity, and chemical and structural tunability.^{6,9–12} Consequently, noncrystalline or amorphous MOFs are attracting attention for applications in, e.g., ion conductivity and molecular encapsulation.¹³

Conventional amorphization of MOFs is achieved through the application of high pressures, thermal treatment, or ball-milling of crystalline frameworks.^{12,13} Recently, the macroscale melting of several zeolitic imidazolate frameworks (ZIFs, including ZIF-4 (cag), ZIF-62 (cag), and ZIF-76 (lta)) has been reported at ca. 650–900 K.^{9,14–17} The melting proceeds on a subnanosecond time frame via dissociation of the Zn–Im (Im = imidazolate, $\text{C}_3\text{H}_3\text{N}_2^-$) bond, before reassociation of the Im with a different Zn center. The energy required to break the Zn–Im bond has been previously calculated as ca. 95 kJ/mol at 840 K for ZIF-4 [$\text{Zn}(\text{Im})_2$].⁹ The high-temperature liquid MOF exhibits a very high viscosity, which suppresses any recrystallization of the glasses formed by melt-quenching.¹⁸ The glasses formed upon cooling possess a continuous random network structure, like that of amorphous SiO_2 .^{9,13}

Among the ZIF glasses known to date, ZIF-62 [$\text{Zn}(\text{Im})_x(\text{bIm})_{2-x}$] (bIm = benzimidazolate, $\text{C}_7\text{H}_5\text{N}_2^-$) is of particular interest because it has the lowest melting temperature.^{16,18} The glass, a_g-ZIF-62, exhibits reversible host–guest adsorption behavior, with gas uptakes of 20.1 (CO_2 , 273 K) and 9.3 (H_2 , 77 K) cm^3 STP g^{-1} at 1 bar.¹⁹ The liquid MOF phase, however, is also of interest as it provides a platform from which materials with specific physical and mechanical properties can be designed and shaped.²⁰ One example of such a material is the MOF crystal–glass composite formed by embedding the as-synthesized phase of MIL-53(Al) (containing unreacted ligand within the pores) within a ZIF-62 glass, which results in stabilization of the high-temperature, open-pore phase of MIL-53(Al) at room temperature.^{21,22}

Currently, there are over 80 000 known MOF structures, although melting behavior has mainly been confined to the ZIF family.^{13,23} Various strategies have been proposed to construct new systems displaying accessible liquid states²⁴ or, alternatively, to lower the melting point (T_m) of existing systems. Such efforts are highly valuable due to (i) the provision of a wider operational temperature window between melting and decomposition, which allows regulation of liquid viscosity, and (ii) the ability to combine a wider range of crystalline

Received: October 29, 2019

Published: January 24, 2020



structures within the liquid state due to the lower temperatures needed for crystal–glass composite processing. Our understanding of the various factors that affect melting remains relatively poor, however, as does our knowledge of the structure, dynamics, and chemical reactivity of the liquid MOF state.

In one of the few studies performed in this area, high-pressure, high-temperature diffraction demonstrated the negative melting-curve behavior of ZIF-62.¹⁹ The structure shows a reduction in melting temperature at high pressures (up to 2 GPa), although related materials with accessible dense phases do not show this behavior.^{19,25} The melting temperature of ZIF-62 can also be regulated via alternating the Im/bIm ligand ratio in the structure.^{16,18}

Coudert and co-workers²⁶ have also investigated the differences in thermal response between ZIF-4 and ZIF-8 [Zn(mIm)₂] (mIm = 2-methylimidazolate, C₄H₅N₂[−]). The experimental *T*_m of ZIF-4 is 863 K, while ZIF-8 decomposes before any possible melting event.^{14,26} Computationally, the *T*_m values of ZIF-4 and ZIF-8 were found to be between 1200 and 1500 K, and >1750 K, respectively. This difference was ascribed to a greater energy barrier for diffusion in ZIF-8, which itself was proposed to be a result of the lack of stabilizing dispersive interactions for mIm linkers in the liquid state (due to the more porous structure of ZIF-8 compared to ZIF-4).

Here, motivated by the prospect of expanding the family of ZIF glasses, realizing new methods to regulate *T*_m and *T*_g (glass transition temperature), and developing a greater understanding of melting behavior and dynamics of the liquid ZIF state, we explore the synthesis of a series of novel glass-forming halogenated ZIF structures. The incorporation of halogenated ligands within MOFs has recently been shown to improve the stiffness of frameworks,²⁷ while they are also capable of introducing permanent dipole moments and facilitating responses to electric fields.²⁸ The electron-withdrawing nature of −F- and −Cl-functionalized ligands also provides an avenue to regulate the melting behavior of the ZIFs, yet this has not been probed. In this work, four novel mixed-ligand ZIFs are constructed using halogenated bIm-based ligands. A combined array of techniques including ¹⁹F solid-state nuclear magnetic resonance (NMR) and in situ terahertz/far-infrared (THz/far-IR) spectroscopy are used to study the effect of halogenation upon melting behavior, as well as the dynamic structure of the resultant liquids. The porosity and mechanical behavior of the novel ZIF glasses formed upon melt-quenching are characterized, demonstrating improvement over their nonhalogenated counterparts.

RESULTS AND DISCUSSION

Crystalline Halogenated ZIF Structures. To produce ZIF glasses with accessible permanent porosity, we selected 5-chlorobenzimidazolate (5-ClbIm, C₇H₄N₂Cl[−]), 5-chloro-2-methylbenzimidazolate (5-Cl-2-mbIm, C₈H₆N₂Cl[−]), 5-fluorobenzimidazolate (5-FbIm, C₇H₄N₂F[−]), and 6-chloro-5-fluorobenzimidazolate (6-Cl-5-FbIm, C₇H₃N₂ClF[−]) linkers, alongside Im (Figure 1a). This followed from a previous report of the use of Im to promote melting and 5-ClbIm as a bulky linker to prevent the collapse of pore channels upon glass formation, allowing the diffusion of guest molecules in and out of the formed quenched glasses.¹⁷ These linkers also provided an array of halogenated functionalities, in order to study the effect on melting behavior.

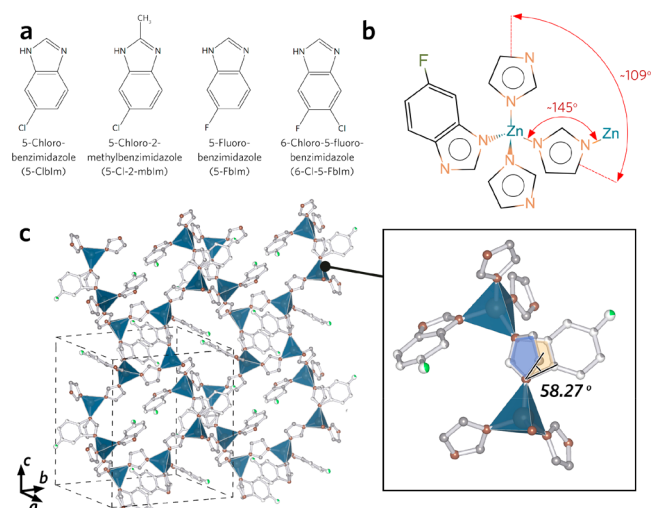


Figure 1. (a) Representations of the halogenated benzimidazolate ligands used in this study. (b) Building unit of ZIF-UC-4 (5-FbIm) with four independent linkers. (c) Representative slice of ZIF-UC-4 (5-FbIm) structure (N, orange; C, gray; ZnN₄ tetrahedra, blue; F, green), showing the six- and four-membered rings. For clarity, only the range of fractional coordinates from 0.3 to 0.8 along *a* axis is shown, and hydrogen and solvent atoms are omitted. The inset highlights a 58.27° orientation deviation of the 5-FbIm linker from the corresponding Im linker position.

Four isostructural frameworks based on the glass-forming ZIF-62 structure were synthesized via a solvothermal process, adapted from that of ZIF-62 (see [Supporting Information](#)).^{17,21} These materials are referred to as ZIF-UC-2 (6-Cl-5-FbIm), ZIF-UC-3 (5-Cl-2-mbIm), ZIF-UC-4 (5-FbIm), and ZIF-UC-5 (5-ClbIm), respectively. UC indicates the university at which these structures were synthesized. The addition of aqueous sodium hydroxide solution, as reported by others,²⁹ was found to be beneficial for the inclusion of bIm-based linkers in ZIF-UC-3–5, due to facile deprotonation in solution.²⁹ After the removal of occluded ligands from the internal pores, single-crystal X-ray diffraction was carried out on suitable crystals. All four frameworks are found to crystallize in the orthorhombic *Pbca* space group with the same 4- and 6-membered ring containing *cag* network topology as ZIF-62 (Figures S1–S13 and Tables S1–S8).³⁰

Like the reported ZIF-62 structure, the asymmetric units of ZIF-UC-2 to ZIF-UC-5 include two independent Zn atoms and four organic linkers (Figure 1b). The preference for structural inclusion of the halogenated ligands, described by their total occupancy factors, follows the order 5-FbIm (ZIF-UC-4) > 5-ClbIm (ZIF-UC-5) > 5-Cl-2-mbIm (ZIF-UC-3) > 6-Cl-5-FbIm (ZIF-UC-2) (Table 1). This tendency is consistent with the trend expected for their steric indexes, δ (the product of the van der Waals volume of the linker and a measure related to size),³¹ where bulkier halogenated ligands have lower probabilities of substituting the Im linker in the framework structure. This is consistent with our previous work.³²

Restrictions on ligand occupancy originate from either the symmetry operations of the specific crystallographic site or from the steric indexes of the halogenated linkers (see [Supporting Information](#)). This leads to a distribution of the halogenated ligands in the four structures, which results in different degrees of disorder. In the first case, the corresponding sites have a maximum occupancy of 50%,

Table 1. Structural Information from Single-Crystal X-ray Diffraction^a

materials	cell parameters (Å)			unit cell volume (Å ³)	halogenated linker total occupancy (%)	solvent occupancy (%)
	<i>a</i>	<i>b</i>	<i>c</i>			
ZIF-UC-2 (6-Cl-5-FbIm)	15.5560(5)	15.6538(5)	18.0875(6)	4404.5(2)	26	
ZIF-UC-3 (5-Cl-2-mbIm)	15.8749(6)	16.2805(6)	18.0681(6)	4669.7(4)	38	78(3)
ZIF-UC-4 (5-FbIm)	15.6970(8)	15.9694(1)	17.9282(2)	4494.1(5)	68	
ZIF-UC-5 (5-ClbIm)	15.7260(8)	16.0184(1)	18.1617(1)	4575.0(5)	57	39(9)

^aFor solvent occupancy, for ZIF-UC-2 (6-Cl-5-FbIm) and ZIF-UC-4 (5-FbIm), the residual electronic density was too low to assign solvent molecules in the asymmetric unit. ZIF-UC-3 and ZIF-UC-4 were refined considering just dimethylformamide (DMF) solvent molecules because this solvent was used in a higher amount in comparison with diethylformamide (DMF) (see refinement details in the Supporting Information).

Table 2. Composition and Thermal Behavior of Halogenated ZIF Crystals and Their Melt-Quenched Glasses

	ZIF-UC-2 (6-Cl-5-FbIm)	ZIF-UC-3 (5-Cl-2-mbIm)	ZIF-UC-4 (5-FbIm)	ZIF-UC-5 (5-ClbIm)	ZIF-62 ¹⁴
crystal precursor (single crystal)	Zn(Im) _{1.87} (6-Cl-5-FbIm) _{0.13}	Zn(Im) _{1.81} (5-Cl-2-mbIm) _{0.19}	Zn(Im) _{1.66} (5-FbIm) _{0.34}	Zn(Im) _{1.72} (5-ClbIm) _{0.28}	
crystal precursor (¹ H liquid NMR)	Zn(Im) _{1.84} (6-Cl-5-FbIm) _{0.16} (DMF) _{0.10}	Zn(Im) _{1.74} (5-Cl-2-mbIm) _{0.26} (DEF) _{0.08}	Zn(Im) _{1.63} (5-FbIm) _{0.37} (DEF) _{0.07}	Zn(Im) _{1.63} (5-ClbIm) _{0.37} (DMF) _{0.07} (DEF) _{0.04}	Zn(Im) _{1.75} (bIm) _{0.25}
melt-quenched glass (¹ H liquid NMR)	Zn(Im) _{1.90} (6-Cl-5-FbIm) _{0.10}	Zn(Im) _{1.75} (5-Cl-2-mbIm) _{0.25}	Zn(Im) _{1.63} (5-FbIm) _{0.37}	Zn(Im) _{1.69} (5-ClbIm) _{0.31}	
melting onset temperature (<i>T</i> _m for crystal)	679 K	663 K	694 K	705 K	673 K
glass transition temperature (<i>T</i> _g for glass)	523 K	609 K	563 K	593 K	591 K
<i>T</i> _g / <i>T</i> _m	0.77	0.92	0.81	0.84	0.88

which prevents symmetry-related bIm-derived linkers coming into contact with each other. In the second case, the increased steric index of bIm-derived linkers relative to Im results in a shift of the halogenated benzimidazolate linker away from the original position to avoid clashing with a neighbor Im linker in a 100% occupied position. This shift results in partial occupancy of the crystallographic sites normally occupied by solvent molecules,³² reducing the occupancy of the halogenated linker due to the competition with solvent molecules.

Therefore, systems in which the total occupancy factor of the halogenated linker is <50% contain just one disordered imidazolate position and only two different configurations for the asymmetric unit (Figure 1c and Figures S1–S13), i.e., ZIF-UC-2 (6-Cl-5-FbIm) and ZIF-UC-3 (5-Cl-2-mbIm) (Table 1). Structures with higher occupancies on the other hand, possess two disordered linker positions. This is the case for ZIF-UC-4 (5-FbIm) and ZIF-UC-5 (5-ClbIm). This results in four possible configurations for the asymmetric unit, creating a higher degree of structural disorder in the system.

The bulk-phase purity of the samples was further confirmed by Pawley refinement of powder X-ray diffraction data (PXRD), using lattice parameters from the determined single-crystal structures. Good agreement between the two confirms the formation of phase-pure samples of halogenated ZIFs (Figure S14). ¹H solution NMR spectroscopy (Figures S15–S18) was carried out by digesting samples of crystalline ZIF-UC-2 to ZIF-UC-5 in a mixture of DCl/D₂O (35%; 0.1 mL) and dimethyl sulfoxide (DMSO-*d*₆) (0.5 mL). The NCHN chemical shifts for Im (~9 ppm region, NCHN_{Im}), the substituted bIm linkers (~9.5 ppm for NCHN_{5-ClbIm}, NCHN_{5-FbIm}, and NCHN_{6-Cl-5-FbIm} and ~2.75 ppm for NCCCH₃N_{5-Cl-2-mbIm}) are well-resolved. Their relative integrations are used to determine the relative ratio of the two linkers in each mixed-linker MOF and, subsequently, the chemical formula for each ZIF structure: ZIF-UC-2 [Zn(Im)_{1.84}(6-Cl-5-

FbIm)_{0.16}·(DMF)_{0.10}], ZIF-UC-3 [Zn(Im)_{1.74}(5-Cl-2-mbIm)_{0.26}·(DEF)_{0.08}], ZIF-UC-4 [Zn(Im)_{1.63}(5-FbIm)_{0.37}·(DEF)_{0.07}], and ZIF-UC-5 [Zn(Im)_{1.63}(5-ClbIm)_{0.37}·(DMF)_{0.07}(DEF)_{0.04}] (Table 2).

Thermal Behavior and Glass Formation. Thermogravimetric analysis (TGA) carried out on the crystalline samples shows the removal of the remaining solvents (*N,N*-dimethylformamide and *N,N*-diethylformamide) at ca. 500 K, followed by thermal decomposition at ca. 730 K for ZIF-UC-2 (6-Cl-5-FbIm), 720 K for ZIF-UC-3 (5-Cl-2-mbIm), and 750 K for ZIF-UC-4 (5-FbIm) and ZIF-UC-5 (5-ClbIm) (Figure 2a). Initial differential scanning calorimetry (DSC) scans on the crystalline samples contain endotherms at ca. 530 K

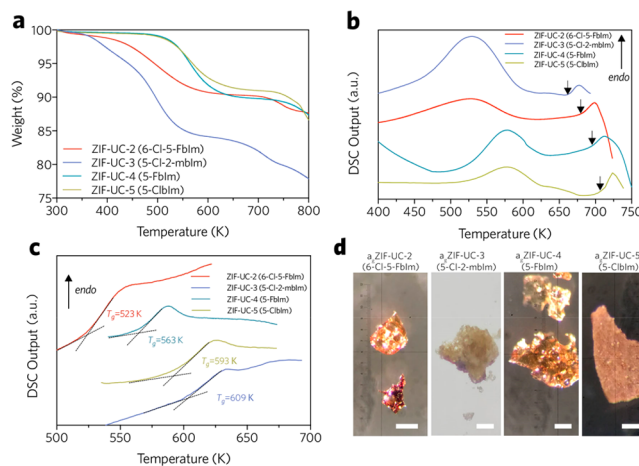


Figure 2. (a) TGA and (b) DSC profiles of the crystalline samples heated at a rate of 20 K min^{−1} under argon. (c) Second up-scan DSC profiles of the melt-quenched glass samples at a rate of 10 K min^{−1}. (d) Microscopic images of the melt-quenched glass samples. The scale bars are 25 μm.

corresponding to the loss of solvent from the frameworks, followed by endotherms ascribed to melting starting at ca. 679 K for ZIF-UC-2 (6-Cl-5-FbIm), 663 K for ZIF-UC-3 (5-Cl-2-mbIm), 694 K for ZIF-UC-4 (5-FbIm), and 705 K for ZIF-UC-5 (5-ClbIm) (Figure 2b and Table 2).

In a separate experiment, to avoid excessive thermal decomposition, crystalline samples were heated above their corresponding melting onset temperatures (693 K for ZIF-UC-2 (6-Cl-5-FbIm), 683 K for ZIF-UC-3 (5-Cl-2-mbIm), 708 K for ZIF-UC-4 (5-FbIm), and 723 K for ZIF-UC-5 (5-ClbIm)) and then cooled back to room temperature. The glasses formed are termed a_g ZIF-UC-2 (6-Cl-5-FbIm), a_g ZIF-UC-3 (5-Cl-2-mbIm), a_g ZIF-UC-4 (5-FbIm), and a_g ZIF-UC-5 (5-ClbIm), with terminology in accordance with previous publications,^{32,33} and were used for subsequent ex situ studies. This was followed by a second DSC heating scan, at 10 K min⁻¹, which identifies T_g values in each case (Figure 2c and Table 2). All melt-quenched glass samples display only diffuse scattering in their PXRD patterns, while optical- and scanning electron microscopy confirm the formation of coherent, continuous, and glassy morphologies upon melt-quenching (Figures 2d, S19, and S20). The Fourier-transform infrared spectroscopy (FTIR) spectra for the crystals and the corresponding melt-quenched glasses show similar features (Figure S21). Because of the relatively close nature of the melting and thermal decomposition temperatures, the melting process was still accompanied by a small quantity of thermal decomposition for two of the crystalline samples. Specifically, ¹H NMR spectroscopy carried out on the glasses (Table 2) confirms a partial loss of the bIm-derived ligand component in a_g ZIF-UC-2 [Zn(Im)_{1.90}(6-Cl-5-FbIm)_{0.10}] and a_g ZIF-UC-5 [Zn(Im)_{1.69}(5-ClbIm)_{0.31}]. However, a_g ZIF-UC-3 [Zn(Im)_{1.74}(5-Cl-2-mbIm)_{0.26}] and a_g ZIF-UC-4 [Zn(Im)_{1.63}(5-FbIm)_{0.37}] are relatively unchanged compared with their crystalline precursors (Figures S22–S25).

The glass transition indicates the transition from solid to liquid-like behavior upon heating. Glasses with similar organic ligand contents possess transition temperatures which correlate with the van der Waals radius of the added side-functional group(s), i.e., a_g ZIF-UC-4 (5-FbIm) (563 K and 1.47 Å) and a_g ZIF-UC-5 (5-ClbIm) (593 K and 1.75 Å). This has also been found previously and is related to larger functional groups increasing the extent of noncovalent interactions within glass or liquid structures.^{14,35} A lower than expected T_g for a_g ZIF-UC-2 (6-Cl-5-FbIm) was observed at 523 K, which was ascribed to the low bIm-derived linker content in the glass phase.

Trends in melting point are less clear. The onsets of melting of the structures in this study, including ZIF-62 itself, are separated by 50 K. The 19 K difference between ZIF-UC-4 (5-FbIm) (T_m = 694 K) and ZIF-UC-5 (5-ClbIm) (T_m = 705 K) is consistent with the greater inductive withdrawal effect of -F compared with -Cl. This experimental observation was further confirmed by density functional theory (DFT) calculations (Figures S26 and S27), which demonstrates an average difference of 2.88 kJ/mol in Zn–N bond energy between Zn(Im)₃(5-FbIm) and Zn(Im)₃(5-ClbIm) tetrahedra.

ZIF-UC-3 (5-Cl-2-mbIm) possesses the lowest T_m observed (663 K), i.e., lower than that of ZIF-62 (T_m = 673 K).^{14,17} The activation free energy, $\Delta F^\ddagger$, for melting has previously been demonstrated to follow the van't Hoff law, $\Delta F^\ddagger(T) = \Delta U^\ddagger - T\Delta S^\ddagger$, where $\Delta U^\ddagger$ and $\Delta S^\ddagger$ are activation energy and entropy, respectively.⁹ From a thermodynamic point of view, the

melting occurs through an activation process, leading to bond breakage at the point of the metastable phase of the ZIF crystal precursor rather than at the limit of stability of the liquid phase.³⁶ The ZIF-UC-3 (5-Cl-2-mbIm) structure contains the most sterically large ligand, and thus the crystal structure formed is the most dense out of the four materials studied here. For dense ZIFs such as this one, it follows that greater dispersive interactions between the framework and the partially dissociated ligand will reduce the energy barrier to melting. For highly porous frameworks on the other hand, a low level of dispersive interactions between framework and detached ligands generates a high activation free energy barrier for melting.²⁶

Glass Structure. The atomic structures of the crystalline ZIFs and their corresponding melt-quenched glasses were probed via synchrotron X-ray total scattering. The lack of crystallinity in the glasses is confirmed by the absence of sharp features in their structure factors (Figures S28–S31). Pair distribution functions (PDFs) for ZIF-UC-4 (5-FbIm) and a_g ZIF-UC-4 (5-FbIm) (Figure 3) were obtained from the total

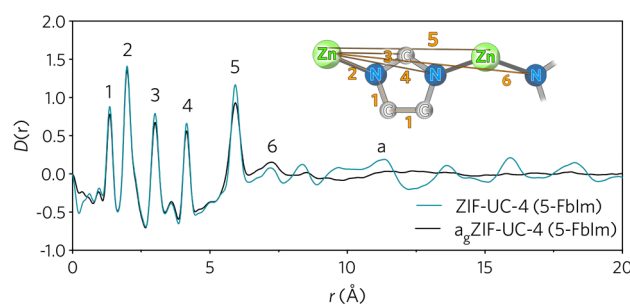


Figure 3. $D(r)$ for ZIF-UC-4 (5-FbIm) and a_g ZIF-UC-4 (5-FbIm) samples. The inset highlights the peak assignment in the structures.

scattering data via appropriate corrections and subsequent Fourier transform (see Supporting Information). As expected, the crystalline samples contain long-range oscillations in the $D(r)$ on length scales above 7 Å, while the glass samples are relatively featureless in this region. For example, the peak in the $D(r)$ for crystalline ZIF-62–5-FbIm at 11.4 Å, marked with an “a”, is associated with the distance between the Zn metal center and the F atom on the next-nearest bIm ligand. This feature is absent in the glass phase, consistent with Zn–N bond breakage and rearrangement upon melting. Generally, below 7 Å, the PDF traces are similar for crystal and glass.¹⁸

Peak assignment was carried out using the PDFGUI software,³⁷ which was used to calculate partial PDFs for the atom pairs in each system. For ZIF-UC-4 (5-FbIm), the first feature at 1.3 Å is assigned to C–C or C–N, and the intense peak at 2 Å is assigned to the Zn–N distance, which confirms Zn–N association in the glass. Correlations between Zn and the second- and third-nearest N atoms of Im ligands can be identified from the PDFs at 5.9 and 7.3 Å (labeled 4 and 6, respectively, in Figure 3). These features are relatively unchanged after vitrification, suggesting the preservation of this short-range order in the glass (Figure 3). Similar PDF results are also obtained for ZIF-UC-2 (6-Cl-5-FbIm), ZIF-UC-3 (5-Cl-2-mbIm), and ZIF-UC-5 (5-ClbIm) (Figures S32–S34).

Figure 4 shows the ¹³C and ¹⁹F magic angle spinning (MAS) NMR spectra of ZIF-UC-4 (5-FbIm) and a_g ZIF-UC-4 (5-FbIm). In the ¹³C MAS NMR spectrum of the crystalline

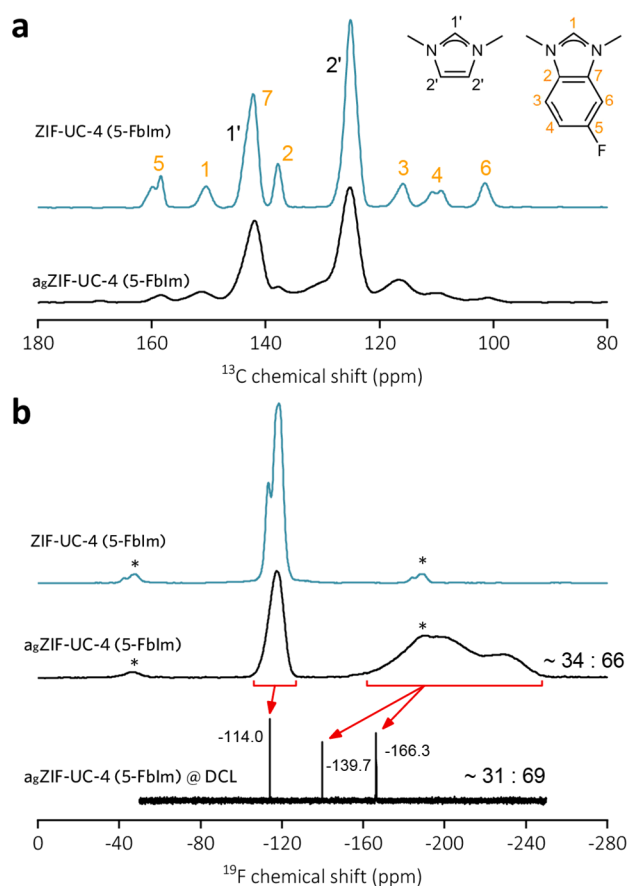


Figure 4. ^{13}C (a) and ^{19}F (b) MAS NMR spectra of ZIF-UC-4 (5-FbIm) and a_Z ZIF-UC-4 (5-FbIm). The spectrum at the bottom of (b) is the ^{19}F solution NMR spectrum of a_Z ZIF-UC-4 (5-FbIm), digested in DCl/DMSO- d_6 . Approximate intensity ratios for the signals at about -114.0 ppm and all the other signals in the ^{19}F spectra of the glass and digested glass are presented. Arrows denote correlations between the ^{19}F NMR signals of the solid glass and the digested glass. Asterisks mark positions of the spinning side bands.

material, the signals of all carbon sites of bIm-derived and Im ligands are clearly resolved. Assignments were performed using chemical shift prediction (ACD/I-lab NMR predictor), ^1H – ^{13}C Lee–Goldburg cross-polarization magic angle spinning (LG-CPMAS), and ^{19}F – ^{13}C CPMAS measurements (Figure S35). The ^{19}F MAS NMR spectrum of the crystalline sample contains one main peak at ca. -115 ppm, which is composed of at least two contributions (Figure 4b). This is explained by the relatively low 5-FbIm/Im ratio present in the material, meaning that two types of Zn centers exist: those that are coordinated to one 5-FbIm and three Im ligands, and those that are surrounded by four Im ligands. Hence, there are pores in the material into which two 5-FbIm ligands protrude (the positions of which are related by symmetry, Figure 1c) and pores into which only one 5-FbIm ligand protrudes. The resulting local environments of 5-FbIm appear sufficiently different to give rise to resolved ^{19}F NMR signals.

The ^{13}C MAS NMR spectrum of a_Z ZIF-UC-4 (5-FbIm) exhibits broader signals at roughly the same isotropic chemical shifts as the spectrum of ZIF-UC-4 (5-FbIm). Broadening of the signals is an expected effect of increased structural disorder and the loss of crystallinity. The ^{19}F MAS NMR spectrum of a_Z ZIF-UC-4 (5-FbIm), however, is more informative. In addition to the signals at about -115 ppm (which are again

slightly broader than the corresponding signals in the crystalline sample), several new broad signals appear between -170 and -240 ppm. NMR measurements show that F nuclei contributing to the narrow signals and F nuclei contributing to the broad signals share the same particles (Figures 4b and S36).

We therefore ascribe the remarkable change in the chemical shift to a considerable change in the nature of the immediate bonding environment of F upon glass formation. ^{19}F – ^{13}C CPMAS NMR measurements confirm this (Figure S37). Selective cross-polarization from ^{19}F nuclei, contributing to the narrow ^{19}F signal of the glass, clearly enhances the signal of C5 C nuclei, i.e., of the nuclei belonging to C atoms within the benzyl ring, which are directly bonded to F. On the contrary, selective cross-polarization from ^{19}F nuclei, contributing to the broad ^{19}F signal of the glass, gives rise to a C spectrum in which only the two signals of Im can be detected. Surprisingly, this observation suggests that F that contributes to the broad, shifted signal is in fact not directly attached to a bIm ligand.

To further verify this observation, the ^{19}F NMR spectrum of a digested sample of a_Z ZIF-UC-4 (5-FbIm) was measured (Figure 4b). The spectrum shows that only a fraction of F nuclei resonate at -114 ppm, i.e., at the chemical shift typical for F of the dissolved 5-FbIm. The spectrum also contains a weak signal at -139.7 ppm and a far stronger signal at -166.3 ppm, which can be consistent with F atoms attached to heavy atoms. On the basis of these observations, we suggest the formation of Zn–F bonds during melting and vitrification. This would explain the observed shift of about -200 ppm in the solid-state ^{19}F MAS NMR spectrum of a_Z ZIF-UC-4 (5-FbIm), which agrees well with the ^{19}F shift in ZnF_2 .³⁸ Other sources of the shifted signal were investigated and ruled out, including the presence of paramagnetic species (through ^{19}F relaxation measurements) and the presence of halogen bonds (through DFT-based calculations of the expected ^{19}F chemical shifts) (Figures S38 and S39). Very similar observations are found also for a_Z ZIF-UC-2 (6-Cl-5-FbIm) (Figure S40). We suspect that the mechanism of dissociation may be similar to thermally activated radical processes in, e.g., halogenated benzene compounds (e.g., chlorobenzene).³⁹ We might also suggest that the fate of the molecule after defluorination is to perhaps form a benzyne intermediate, allowing further reaction with imidazole to form heterocyclic structures or, alternatively, to form *N*-alkyl-*N'*-arylimidazolium salts. This, however, is far beyond the scope of this work and investigations would be necessary.^{40–42}

THz/Far-IR Study of Bonding Environments. While the midrange IR region is usually associated with fundamental molecular vibrations, THz/far-IR spectra can reveal spectral features associated with quasi-localized and collective framework dynamic properties, which provides an effective probe to study dynamic framework properties upon glass formation.⁴³ Synchrotron THz/far-IR absorption spectroscopy has been shown to be useful in determining the spectra of dynamic vibrational motions up to ~ 700 cm^{-1} and ~ 21 THz. The spectra above this region are not shown because of the cutoff by the beam splitter used for this beamline instrument.

Compared to the ZIF precursors ($\text{Zn}(\text{NO}_3)_2$, HIm, and 5-F-1-HbIm, Figure S41), there are three main sources of low-energy vibrations of crystalline ZIF-UC-4 (5-FbIm) within the spectral region below 700 cm^{-1} (Figure 5): in-plane and out-of-plane aromatic ring deformations (ca. 630 – 700 cm^{-1}), Zn–N stretching movements (ca. 280 – 330 cm^{-1}), and 4-

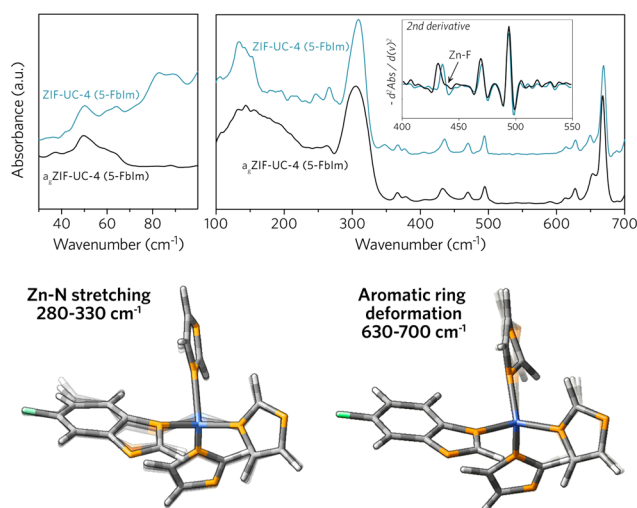


Figure 5. THz/far-IR spectra of ZIF-UC-4 (5-FbIm) and a_Z ZIF-UC-4 (5-FbIm) with the inset being the second derivative of the spectra highlighting the emergence of a shoulder peak at ~ 440 cm^{-1} after melt-quenching, and a schematic diagram of deformation modes of ZnIm_3 5-FbIm tetrahedra and aromatic rings used for DFT calculations. Zn, blue; N, orange; F, green; C, gray; and H, white.

membered and 6-membered ring collective motions (ca. 30–200 cm^{-1}).⁴⁴ The experimental spectrum shows similarities with the theoretical DFT calculations on the $\text{Zn}(\text{Im})_3$ (5-FbIm) tetrahedron (150–700 cm^{-1} , Figure S42). It should be emphasized that, for the low-wavenumber region, it is difficult to establish a good agreement between experimental and calculated data because of the complexity of the collective framework vibrations (23–100 cm^{-1} , Figure S43).⁴⁵

For the melt-quenched glass phase, a clear peak broadening compared with the crystal precursor is observed, especially in the region of 23–200 cm^{-1} (Figure 5), which has been linked to the collective movement of gate-opening and breathing effects involving 4- and 6-membered rings.⁴⁴ Similar behavior is also observed for crystalline and glass ZIF-UC-2 (6-Cl-5-FbIm) (Figures S44 and S45), ZIF-UC-3 (5-Cl-2-mbIm) (Figures S46 and S47), ZIF-UC-5 (5-ClbIm) (Figures S48 and S49), and ZIF-62 (Figures S50 and S51). In addition, the presence of a broad peak centered at ca. 440 cm^{-1} , as suggested by the second derivative spectra in the inset, is consistent with the emergence of a Zn–F bond after vitrification.⁴⁶ DFT calculations on a $\text{Zn}(\text{Im})_3$ (F) tetrahedron confirm this, with good agreement in the peak position (ca. 420 cm^{-1} , Figure S52). The emergence of a broad peak (ca. 460 cm^{-1}) assigned to Zn–F is also observed for ZIF-UC-2 (6-Cl-5-FbIm) after vitrification (Figure S44), while this broad peak is (evidently) absent for the ZIFs that do not contain the fluoro moiety: ZIF-UC-3 (5-Cl-2-mbIm), ZIF-UC-5 (5-ClbIm), and ZIF-62 (Figures S46, S48, and S50).

Dynamic Properties of Crystal, Glass, and Liquid. The relatively low glass transition temperature of the halogenated ZIF glasses opens opportunities to probe vibrational properties and low-frequency conformational dynamics in the liquid and glass phases and compare these to those present in the crystalline solids. In situ heating experiments up to 573 K were performed at the Australian Synchrotron THz/far-IR beamline. An in-house built gas purge-box was used to maintain an argon environment during the heating process (Figure S53). Because of the absence of thermal hysteresis for the THz/far-IR spectra

during heating and subsequent cooling steps (Figure S54), only the heating cycle was investigated.

Heating of ZIF-UC-4 (5-FbIm) and a_Z ZIF-UC-4 (5-FbIm) to 573 K does not lead to a significant change in peak shape, with only a redshift in the peak positions for Zn–N tetrahedra observed (ca. 310–300 and 305–290 cm^{-1} , Figures 6a and b

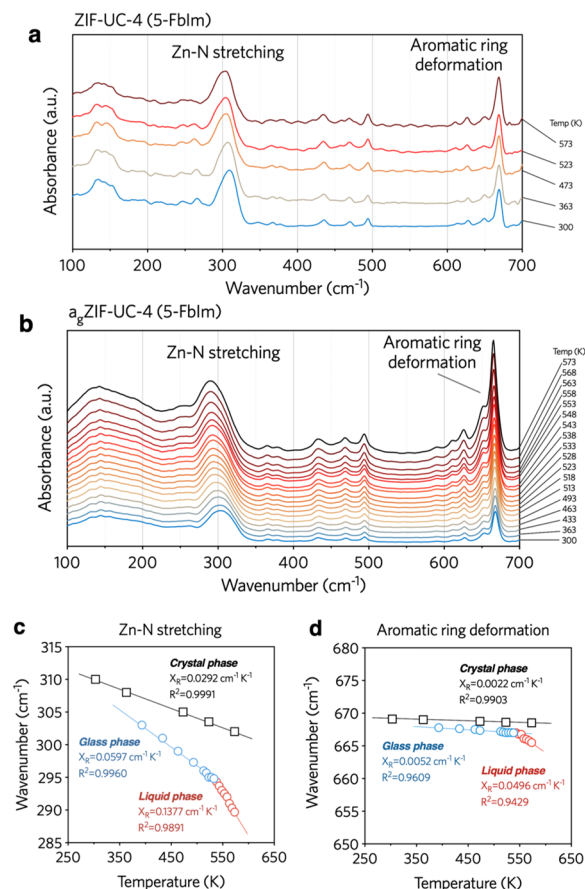


Figure 6. Synchrotron THz/far-IR spectra during the heating of ZIF-UC-4 (5-FbIm) and a_Z ZIF-UC-4 (5-FbIm). Temperature-resolved in situ THz/far-IR spectra for (a) ZIF-UC-4 (5-FbIm) and (b) a_Z ZIF-UC-4 (5-FbIm) up to 573 K. (c, d) Peak position shifts for the Zn–N stretching and aromatic ring deformation modes, respectively (see text for a detailed explanation of the graphs).

and S55). Ryder et al. previously demonstrated that the quasi-delocalized ZnN_4 vibrational modes obey the Bose–Einstein distribution,^{47,48}

$$\nu(T) = \nu_0 - \frac{X_R T_C}{\exp\left(\frac{T_C}{T}\right) - 1} \text{ where } T_C = \frac{h\nu_0}{k_B}$$

where ν_0 is the frequency of the vibration at 0 K and T_C is the vibrational temperature (K), which is obtained from statistical thermodynamics via the Planck constant, h , and the Boltzmann constant, k_B . In particular, they defined the linear curve fitting slope, X_R , as the spectral position shift coefficient. X_R was demonstrated to be correlated to the thermally induced deformation of the ZnN_4 environment and, thus, affected by the exact nature of the coordinative environment. While a detailed quantum mechanical analysis of this Bosonic behavior lies beyond the scope of this Article, we applied the same logic here to deduce values of X_R for all crystalline and glass phases.

Compared with the crystal precursor, a clear increase of the X_R value of Zn–N stretching for the glass phase is observed (Figure 6c). This is followed by a more significant enhancement above the T_g (563 K) for the liquid phase, which demonstrates the greater flexibility of the Zn environment in the liquid phase.

As expected, negligible changes are observed in the deformation-related behavior of the aromatic ring within crystalline ZIF-UC-4 (5-FbIm) upon heating (Figure 6d). The value of X_R is a magnitude smaller than that for the Zn–N bond stretching—attributed to the rigidity of the aromatic ring.⁴⁹ However, this same mode in a_g ZIF-UC-4 (5-FbIm) displays a substantial redshift when the temperature approaches T_g (ca. 563 K, Figure 6d). For example, while the X_R for Zn–N of a_g ZIF-UC-4 (5-FbIm) increases by 1.3 times above the T_g , the a_g ZIF-UC-4 (5-FbIm) X_R for aromatic ring deformation increases to nearly 10 times the original value (0.0052 – 0.0496 $\text{cm}^{-1} \text{K}^{-1}$, Figure 6d) upon heating above T_g .

To further confirm the phenomenon, we conducted in situ THz/far-IR heating tests for all halogenated ZIFs—ZIF-UC-2 (6-Cl-5-FbIm), Figures S56–S58; ZIF-UC-3 (5-Cl-2-mbIm), Figures S59–S61; ZIF-UC-5 (5-ClbIm), Figures S62–S64, together with ZIF-62 (Figures S65–S67)—and observed similar behavior for all crystals and glasses. The T_g of a_g ZIF-UC-3 (5-Cl-2-mbIm), a_g ZIF-UC-5 (5-ClbIm), and a_g ZIF-62 are all above 573 K, i.e., the maximum temperature of this experiment, and thus, little variance in X_R values is noted.²¹ However, similar changes in X_R above the T_g of ca. 523 K are observed for a_g ZIF-UC-2 (6-Cl-5-FbIm). Interestingly, the Zn–N stretching X_R for different ZIF crystals follows the order of ZIF-UC-3 (5-Cl-2-mbIm) (0.0374 $\text{cm}^{-1} \text{K}^{-1}$) > ZIF-62 (0.0362 $\text{cm}^{-1} \text{K}^{-1}$) > ZIF-UC-2 (6-Cl-5-FbIm) (0.0343 $\text{cm}^{-1} \text{K}^{-1}$) > ZIF-UC-4 (5-FbIm) (0.0292 $\text{cm}^{-1} \text{K}^{-1}$) > ZIF-UC-5 (5-ClbIm) (0.0255 $\text{cm}^{-1} \text{K}^{-1}$), which shows excellent agreement with their melting onset temperatures (Figure 7 and Table S9).

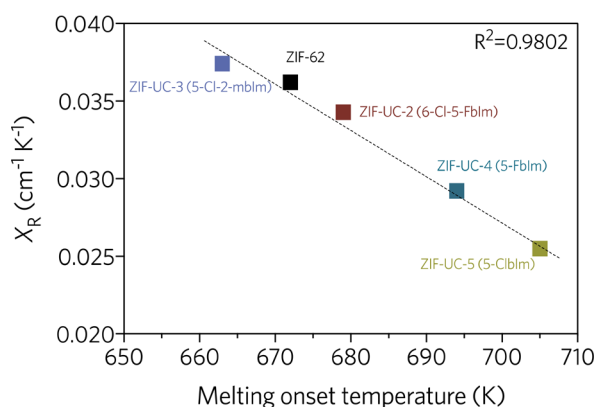


Figure 7. Correlation between the melting onset temperature and X_R .

The decrease in melting temperature with greater values of X_R is consistent with the computationally calculated melting mechanism for ZIFs. For example, melting has been theorized to proceed via metal–linker bond breaking at a critical temperature, where the ratio of the thermally induced vibrational amplitude of metal-linked atoms (i.e., N) and the M–N bond distance reach a critical point, i.e., Lindemann's law.⁹ Since X_R describes the increasing deformation of the ZnN_4 tetrahedra, it must also be correlated to both bond

distance and the thermal vibration of the ligand. We also observe that the melting tendency has no clear correlation with the Zn–N bond strength at room temperature (Figure S68).

Porosity. ZIF-UC-2 (6-Cl-5-FbIm), ZIF-UC-3 (5-Cl-2-mbIm), ZIF-UC-4 (5-FbIm), and ZIF-UC-5 (5-ClbIm) all display a low N_2 uptake at 77 K (Figure S69) with relatively low level of Brunauer–Emmett–Teller (BET) surface areas, in the range of ca. 7.10 – 86.73 m^2/g .⁵⁰ These values are consistent with the data reported in the literature, where the relatively dense structure of ZIF-62 blocks the entry and exit of N_2 gas molecules into its inner framework structure.⁵¹ On the other hand, smaller gas molecules like CO_2 can diffuse into the crystalline framework, and the absence of adsorption/desorption hysteresis indicates the rapid adsorptive/desorptive kinetics for these crystals (measured at 273 K, Figure S70). Our earlier work demonstrates that a reduction in porosity, and thus gas adsorption capacity, is to be expected after vitrification of a crystalline ZIF.¹⁹ A substantial degree of microporosity can be preserved, however, because of the presence of bulky bIm-based ligands, which prevents the total collapse of the porous interior upon liquid formation.^{11,17} CO_2 adsorption isotherms at 273 K for the melt-quenched glass samples confirm this is also the case here (Figure 8a): all melt-quenched ZIF glasses show reversible adsorption of CO_2 , suggesting the relatively rapid diffusion of guest molecules within these microporous structures.

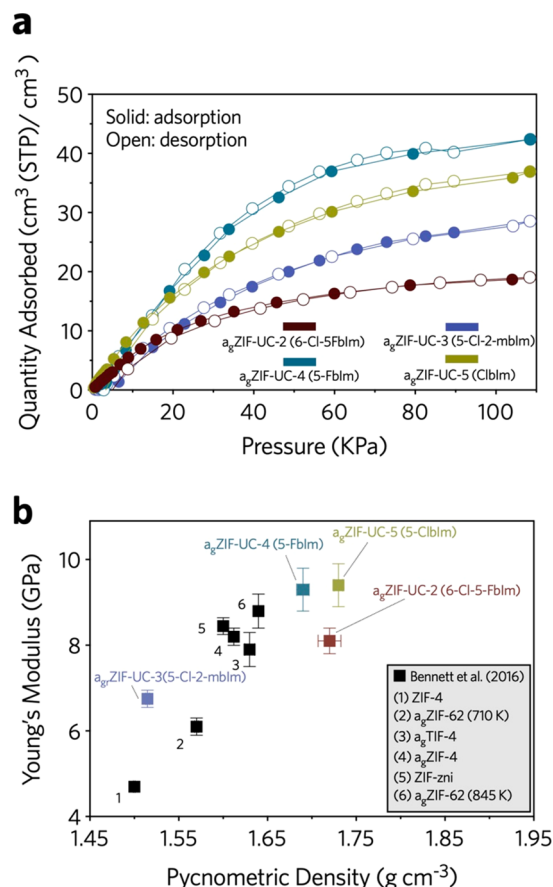


Figure 8. (a) CO_2 gas adsorption/desorption isotherm for melt-quenched ZIF glasses. (b) Plot of pycnometric density against Young's modulus of existing MOF glasses in this work, alongside the materials reported in this work.¹⁴

The incorporation of fluorinated ligands can improve the CO₂ uptake for both crystalline and glass materials. For example, despite ZIF-UC-4 (5-FbIm) and ZIF-UC-5 (5-ClbIm) exhibiting comparable quantities of bulky bIm-based ligands (Table 2), ZIF-UC-4 (5-FbIm) exhibits a markedly higher CO₂ uptake that is rationalized in terms of the stronger interaction between CO₂ guests and framework F atoms in the narrow channels of the MOF.⁵² Similar behavior of other F-functionalized MOFs has been reported by other researchers.^{53–55} After melting, a_gZIF-UC-4 (5-FbIm) displays a CO₂ uptake of 42 cm³ (STP)/cm³ at 1 bar, which is ca. 15% higher than that for a_gZIF-UC-5 (5-ClbIm) and ca. 30% higher than that for a_gZIF-62.¹⁹

Mechanical Properties. Nanoindentation has been previously applied to investigate the mechanical properties including Young's moduli (*E*) and hardness of crystalline and amorphous MOFs.^{14,21} The average Young's modulus was recorded from load-displacement data (200–1000 nm) on polished glass samples (Figures S71–S74). Both a_gZIF-UC-4 (5-FbIm) and a_gZIF-UC-5 (5-ClbIm) show comparable values of *E* ≈ 9.4 and 9.3 GPa, respectively, placing them roughly at the upper bound of the existing ZIF glasses (Figures 8b and S75). The mechanical properties of MOF materials do not always follow density–stiffness rules but are determined more by the primary (metal–organic coordination structure) and secondary (nonbonded interactions) networks, which provide extra support for the structural weak points.⁵⁶ Consistent with these reports, we postulate that the presence of halogenated moieties on the bIm-derived ligands can introduce more effective secondary network structures via nonbonded interactions, or indeed by the Zn–F bonds, which raise *E* in these cases above that found for a_gZIF-62 (*E* ≈ 6.1–8.8 GPa, Figure 8b). Attempts have also been made to prepare bulk halogenated glass samples by first pelletizing samples of a_gZIF-UC-3 (5-Cl-2-mbIm), before annealing above the glass *T*_g.⁵⁷ The resultant bulk glass (termed as a_gZIF-UC-3 (5-Cl-2-mbIm)) has a dark brown, opaque appearance, with a diameter of 13 mm (Figure S76) and a Young's modulus of 6.66 ± 0.14 GPa.

CONCLUSION

Four new crystalline ZIFs, ZIF-UC-2 (6-Cl-5-FbIm), ZIF-UC-3 (5-Cl-2-mbIm), ZIF-UC-4 (5-FbIm), and ZIF-UC-5 (5-ClbIm) have been reported and have all been shown to undergo melting. The melting of ZIFs has previously been reported to be affected by the crystal structure of the precursor, the stoichiometric ratio of different ligands, and the type of metal node.^{16,18} The array of experimental techniques provided here show that the presence of electron-withdrawing moieties on the benzimidazolate ligand may weaken the Zn–N bond and also lead to a small reduction in the melting temperature of the ZIF crystal. The results also show that the presence of bulkier ligands results in a lower free activation energy through dispersive interaction-based stabilization of dissociated ligands in the liquid phase. These findings are encapsulated by in situ THz/far-IR studies that reveal the connection between Zn–N bond motion and melting behavior. Specifically, they provide evidence of a correlation between the ZIF melting temperature and the propensity of thermal-induced Zn–N polyhedral deformation, which provides experimental evidence of a melting mechanism being underpinned by the thermally induced vibrational amplitude of

the N atoms reaching a critical point, which leads to the breaking of the Zn–N bond.

In addition, solid-state NMR results provide specific evidence for the formation of Zn–F bonding in the glass phase, implying a reduction in the overall network connectivity. The extent of this rearrangement process is dependent upon treatment temperature, although the fate of the proportion of fluorobenzimidazolate ligands from which F detaches is beyond the scope of this work. The presence of a fluorinated moiety within the glass of ZIF-UC-4 (5-FbIm) also notably improves gas uptake for CO₂, compared with ZIF-62 and ZIF-UC-5 (5-ClbIm) containing chlorinated benzimidazolate ligands.

The results help explain the glass-forming ability (GFA) of ZIF materials, which were analyzed using their respective *T*_g/*T*_m ratios (Table 2). This ratio for conventional metallic, covalent, and ionic melt-quenched glasses follows the empirical Kauzmann “2/3 Law”³⁴ and has been used previously to suggest the resistance to recrystallization of the supercooled liquid state.¹⁸ The highest GFA (0.92) for the samples studied here is observed for ZIF-UC-3 (5-Cl-2-mbIm). This is ascribed to the presence of the sterically bulky 5-Cl-2-mbIm ligand, which suppresses the ones for diffusion in the liquid phase. This is higher than the reported GFA for ZIF-62 (0.88). At the same time, slightly lower GFAs for the ZIF-UC-2 (0.77) and ZIF-UC-4 (0.81) are attributed to the formation of Zn–F bonds, which breaks the continuity of the random network and improves the diffusivity of the ligands in ZIF liquid, thus lowering *T*_g. The GFAs for the halogenated ZIFs are still higher than ionic, metallic, and covalent glasses.

Importantly, together with recent progress on lowering the melting temperature of ZIFs through (i) adjustment of organic linker ratio and (ii) inclusion of multiple organic linkers and (iii) cobalt, these results may be used to further lower the melting temperature of ZIFs. Such research is essential if industrially viable working temperatures are to be achieved and larger-scale studies on the applications of such glasses are to be performed.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/jacs.9b11639>.

Crystallographic data for ZIF-UC-2 (ClFbIm) (CIF)

Crystallographic data for ZIF-UC-3 (ClmbIm) (CIF)

Crystallographic data for ZIF-UC-4 (FbIm) (CIF)

Crystallographic data for ZIF-UC-5 (ClbIm) (CIF)

Experimental and characterization procedures, data on crystal structure, PXRD and Pawley refinement, liquid-state NMR, solid-state NMR, DFT calculation for charge density distribution, SEM image, FTIR, X-ray total scattering, pair distribution function, synchrotron THz/farIR spectra, gas sorption, and nanoindentation and density (PDF)

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Notes

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ACKNOWLEDGMENTS

T.D.B. thanks the Royal Society for a University Research Fellowship (UF150021) and a research grant (RSG\R1\180395). J.H. and T.D.B. also thank EPSRC (EP/

R015481/1). J.H. thanks the Australian Research Council for a DECRA Fellowship (DE190100803). V.C. acknowledges financial support (DP180103874). S.L. acknowledges the China Scholarship Council (CSC). A.F.S. acknowledges the EPSRC for a Ph.D. studentship under the industrial CASE scheme along with Johnson Matthey PLC (JM11106). A.M.B. acknowledges the Royal Society for funding (RGF\EA\180092) as well as the Cambridge Trust for a Vice Chancellor's Award (304253100). Z.D. thanks the Cambridge Trust and the China Scholarship Council. The calculations were performed at the U.K. National Supercomputing Service, ARCHER. Access was obtained via the UKCP consortium and funded by EPSRC under Grant no. EP/P022596/1. A.K. and G.M. acknowledge financial support from the Slovenian Research Agency (research core funding no. P1-0021 and research project Z1-9171). We thank U. Javornik for help with the liquid-state ^{19}F NMR measurements. Generous computational resources are provided (to B.C.) by RIKEN Information Systems Division (Project Q19266). The authors acknowledge the facilities, and the scientific and technical assistance, of the Australian Microscopy & Microanalysis Research Facility at the Centre for Microscopy and Microanalysis, The University of Queensland. Part of this research was undertaken on the THz/Far-IR beamline at the Australian Synchrotron, part of ANSTO (M14790 and EU15604). We acknowledge the provision of synchrotron access to Beamline I15-1 (EE20038-1 and EE20038-2) at the Diamond Light Source.

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